Thermodynamics of Inequalities: From Precariousness to Economic Stratification

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Abstract

Increasingly large social and economic inequalities are observed throughout the world. Theorists after Pareto have studied this phenomenon in terms of the tail structure of the wealth distribution at a given time. Unfortunately, this approach leaves unaddressed the *dynamics* of inequalities in non-equilibrium situations, e.g. under redistribution policies. Here we introduce a thermodynamical theory of inequalities based on the analogy between economic stratification and statistical entropy. Within this framework we identify the combination of *upward mobility* with *precariousness* as a fundamental driver of inequality. We formalize this statement by a "second-law" inequality displaying upward mobility and precariousness as thermodynamic conjugage variables. Our method can be generalized to gain insight into the dynamics of inequalities in any Markovian model of socioeconomic interactions.

Keyword: economic inequalities, stratification, precariousness, stochastic wealth models, Markov process, fluctuation theorem, entropy

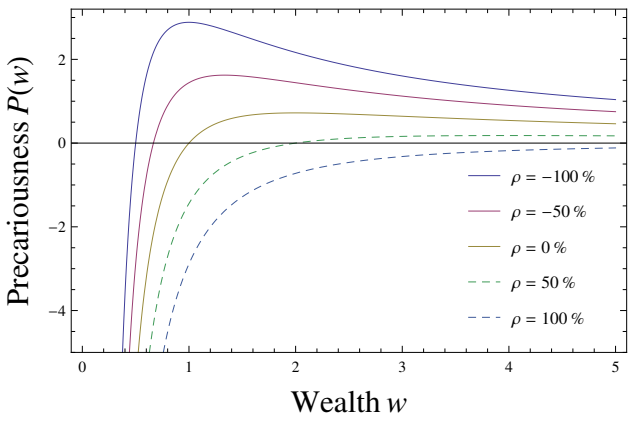


Figure1: "Precariousness" function in a model of stochastic wealth for various values of the mean effective interest rate ρ. The product of precariousness P and mean net gain Δw is a lower bound on inequality growth in sociodynamics, for the same reason that the product of inverse temperature β and heat Q is a lower bound on entropy growth in thermodynamics.